

Fake News Detection

Data Science Final Project



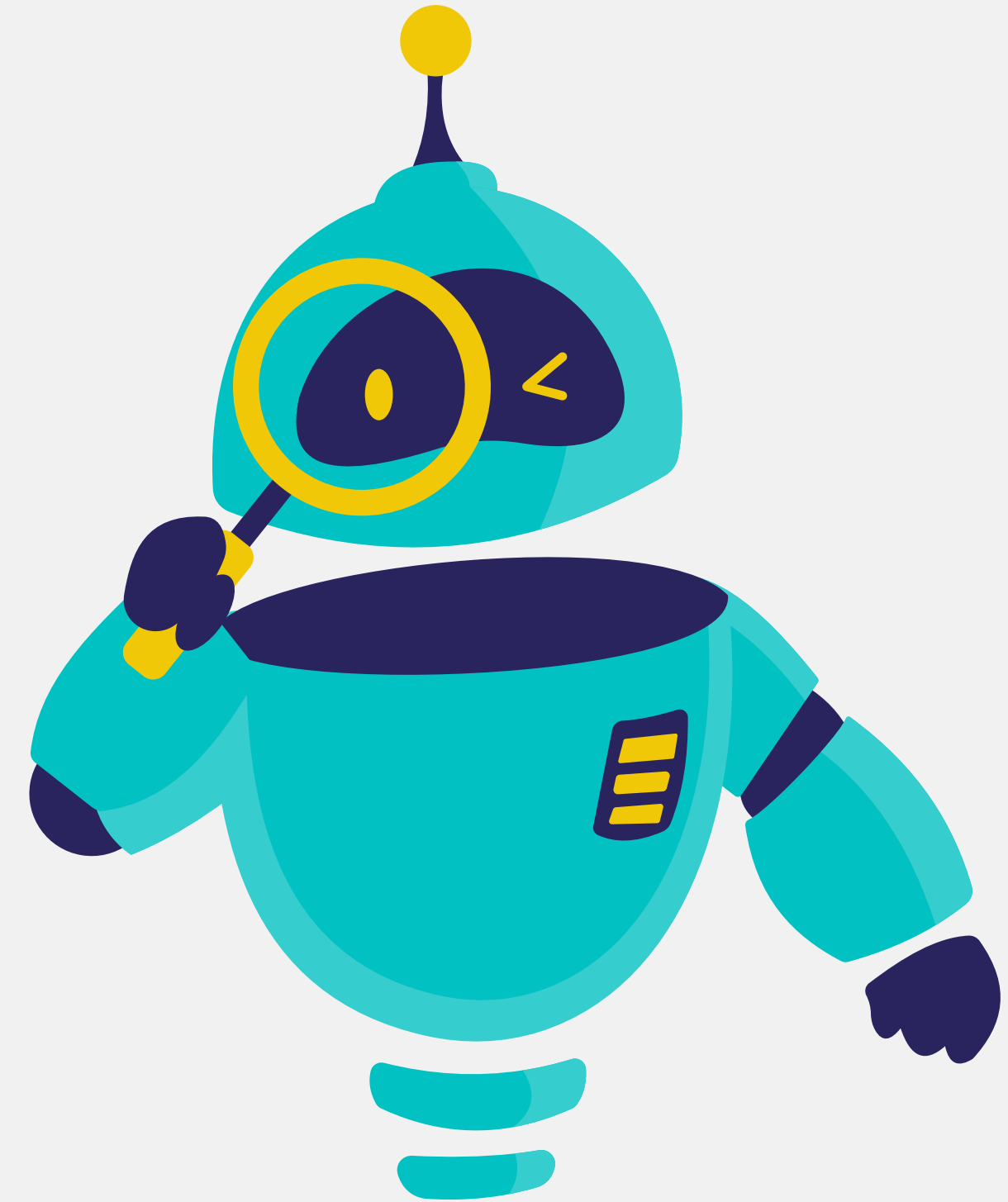
University of Tehran
Spring 2025

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What is our Task About?

A **classification task** for detecting fake news from real ones. In an age where misinformation spreads faster than truth, uncovering the subtle patterns of fake news becomes crucial.

On this project, we explored how fake and true news differ across time, titles, language, and topics, by revealing the impact of political events, platform policies, and linguistic choices. Then we deployed a classifier model for detecting fake news.





About our Dataset

 **Source:** Kaggle

 **Structure:** *Two CSV files*

True.csv → 21,418 news

Fake.csv → 23,480 news

→ ***Totally 44,898 news***



Title

The headline of the news article.
Often crafted to grab attention
and can reflect the emotional
tone or intent of the content.

Text

The full body of the article
containing the main narrative.
This section provides the context,
facts, or misinformation that
supports the headline.

Subject

Indicates the category of the news,
such as World News or Politics.
It helps identify patterns in fake vs.
true news distribution across
topics.

Date

Shows the publication date of each
article.
Useful for analyzing time-based trends
and the impact of political events on
news volume.

Data Info

Among the four key columns, only the date column contains missing values, with 11,613 null entries, indicating that about 25.8% of the news items lack a publication date.

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 44898 entries, 0 to 44897
Data columns (total 18 columns):
#   Column              Non-Null Count  Dtype
---  -
0   title                44898 non-null  object
1   text                 44898 non-null  object
2   subject              44898 non-null  object
3   date                 33285 non-null  object
4   label                44898 non-null  object
```

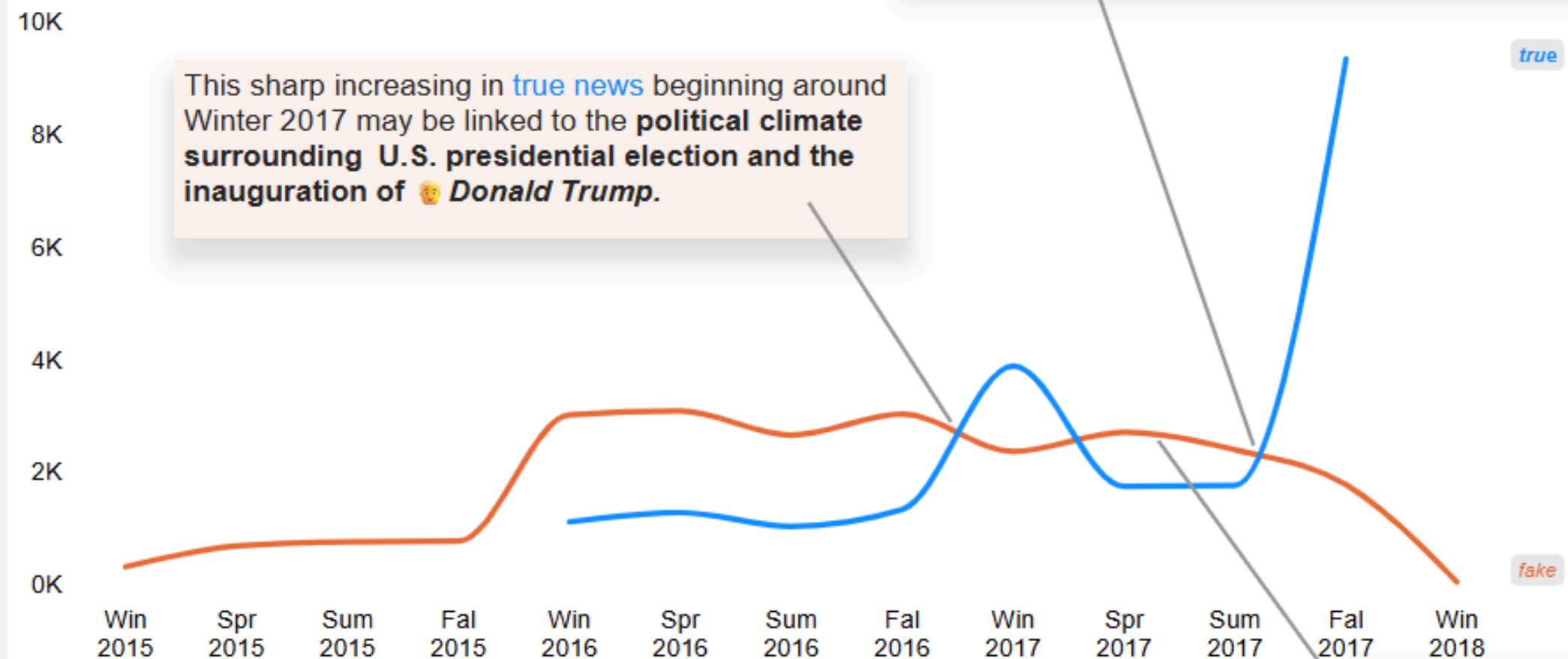
A sample of our data

title	text	subject	date	label
Pope Francis Just Called Out Donald Trump Dur...	Pope Francis used his annual Christmas Day mes...	News	2017-12-25	fake
...	...	...	...	...
'Fully committed' NATO backs new U.S. approach...	BRUSSELS (Reuters) - NATO allies on Tuesday we...	worldnews	2017-08-22	4 true



EDA using PowerBI

News publication trend over time



This sharp increasing in **true news** beginning around Winter 2017 may be linked to the **political climate** surrounding U.S. presidential election and the inauguration of 🗣️ **Donald Trump**.

Since Summer 2017, the publication of **true news** has shown a steady upward trend. **This shift may be attributed to actions taken by major digital platforms** which began implementing policies to limit the spread of **fake news**.

 **+254.1%**
True news growth after 2017

Meanwhile, **fake news**, which was more prevalent earlier, begins to **decline** steadily after 2017, **suggesting the potential effectiveness of platform-level intervention and algorithmic moderation**.

After 2017, the surge in **true news** and decline in **fake news** suggest that political shifts and platform level moderation significantly reshaped the digital news landscape.

EDA using PowerBI

True/Fake	Mon	Tue	Wed	Thu	Fri	Sat	Sun
fake	3308	3526	3557	3582	3479	2947	3068

True/Fake	Mon	Tue	Wed	Thu	Fri	Sat	Sun
true	3064	3749	4184	4106	3695	1192	1427

Fake news peaks early in the week and stays steady, with less decline on weekends, likely because creators target times when people are more online and have more free time. In contrast, **true news** drops sharply on weekends since it's mainly produced by professional 🏢 agencies that work on regular weekdays.

EDA using PowerBI

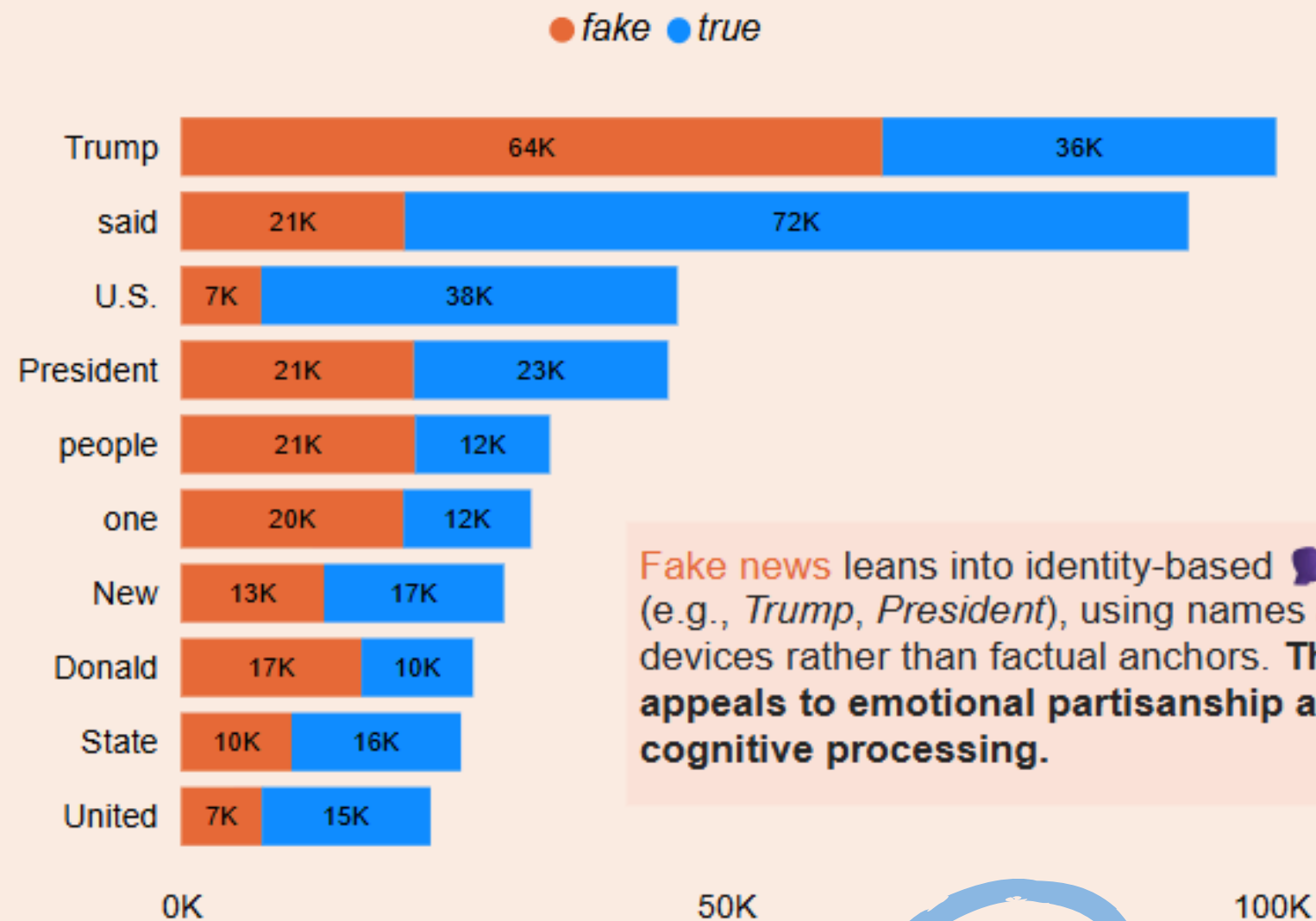
Distribution of capitalized words in news titles



Titles of **fake news** tend to use more **ABCD** capitalized words, possibly to grab attention or evoke urgency. In contrast, **true news** titles are generally more balanced and restrained.

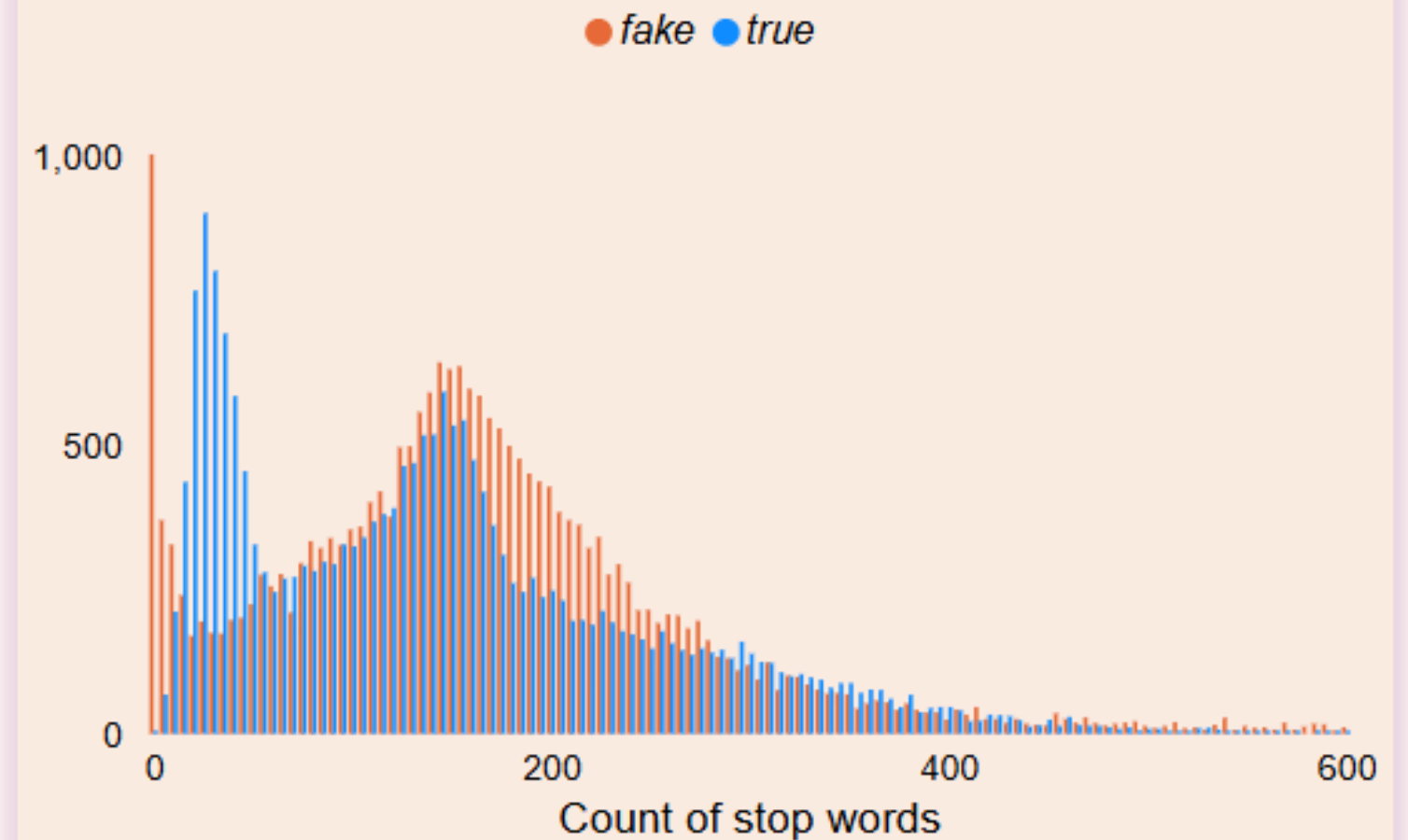
EDA using PowerBI

Top 10 frequent words used in news



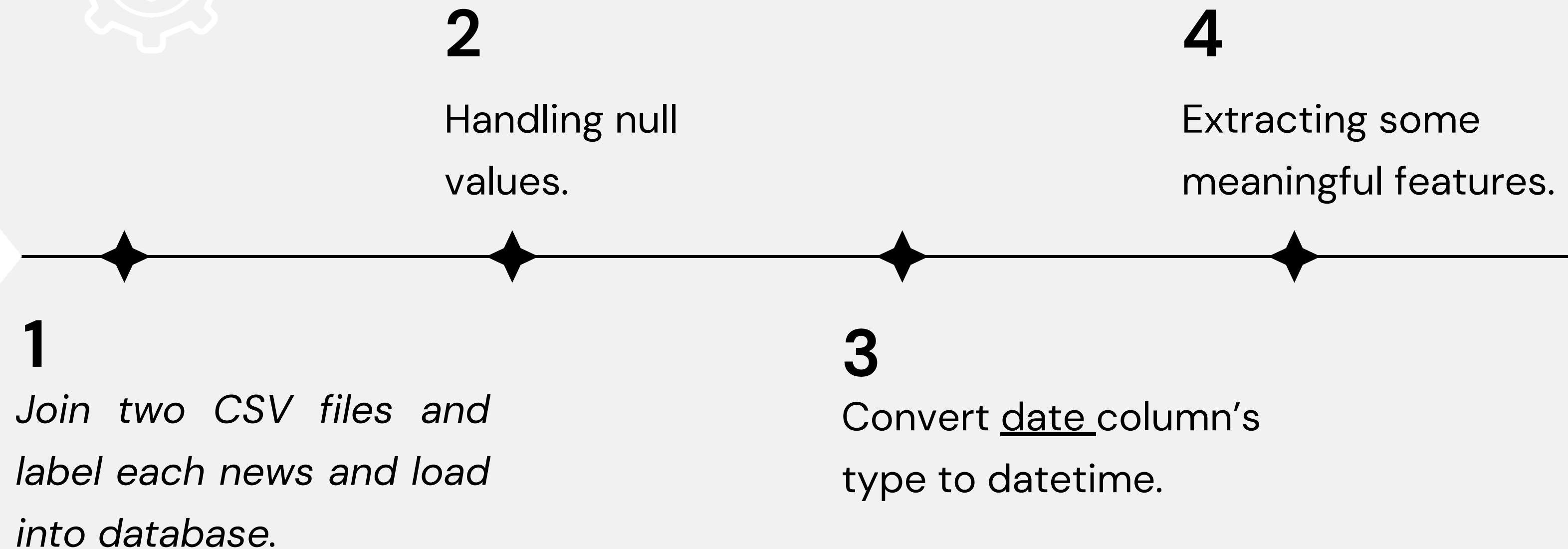
Fake news leans into identity-based vocabulary (e.g., *Trump*, *President*), using names as rhetorical devices rather than factual anchors. **This approach appeals to emotional partisanship and fast cognitive processing.**

Distribution of stop words in news text

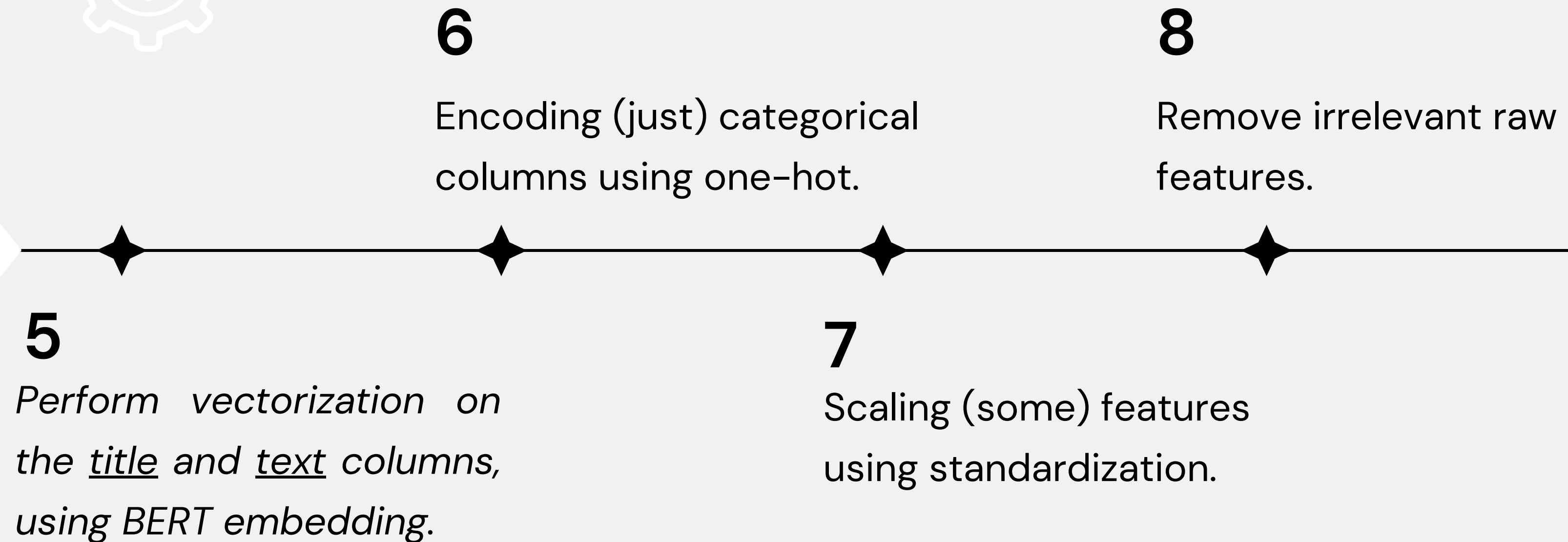


True news tends to include more stop words. This reflects more natural, formal writing styles commonly found in journalistic standards. But **fake news**, may use fewer of these words, focusing on brevity and emotional impact.

Preprocessing & Feature Engineering Steps



Preprocessing & Feature Engineering Steps



Preprocessing & Feature Engineering Steps

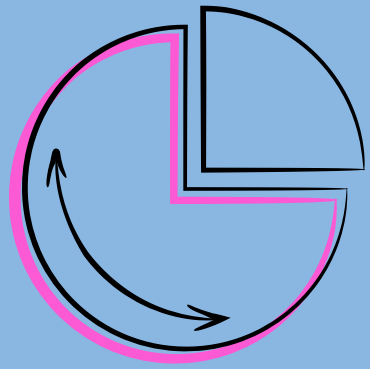
10

Remove highly correlated features. ($> 90\%$)

At the end, load the new data into database.

9

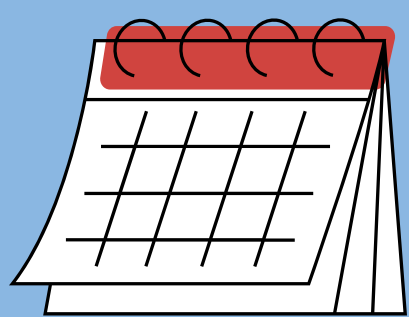
Delete noisy (duplicate, low-variance, sparse) features.



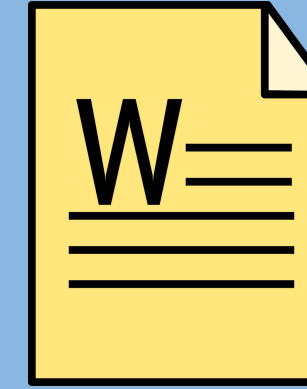
"avg_word_length"
"stopword_ratio"



"lexical_diversity"
"word_density"
"emotional_density"



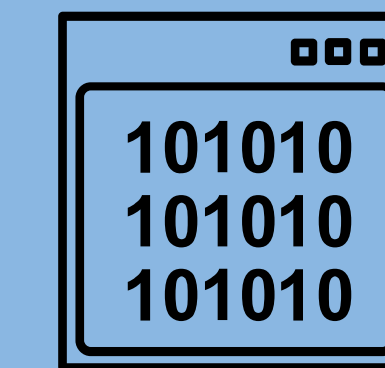
"publish_day"
"publish_month"
"season"



"count_capital_words"
"count_emotional_words"
"count_stopwords"



"number_count"
"word_count"
"sentence_count"



"is_question"
"is_weekend"

Extracted Features



Models used

Now it's time to deploy our model! First we splitted our dataset into *train*(80%), *validation*(10%) and *test*(10%).

We used these traditional models:

- **XGBoost** → boosting
- **CatBoost** → categorical
- **LightGBM** → leaf-wise

Also these deep learning models:

- **Extra Trees** → random forest
- **HistGB** → continuous
- **Multi-Layer Perceptron (MLP)** → non-linear relations

Models Evaluation Results on Validation Set

XGBoost best model evaluating...

	precision	recall	f1-score	support
0	1.00	0.99	0.99	2348
1	0.99	0.99	0.99	2142
accuracy			0.99	4490
macro avg	0.99	0.99	0.99	4490
weighted avg	0.99	0.99	0.99	4490

Models Evaluation Results on Validation Set

CatBoost best model evaluating...

	precision	recall	f1-score	support
0	1.00	1.00	1.00	2348
1	1.00	1.00	1.00	2142
accuracy			1.00	4490
macro avg	1.00	1.00	1.00	4490
weighted avg	1.00	1.00	1.00	4490

Models Evaluation Results on Validation Set

LightGBM best model evaluating...

	precision	recall	f1-score	support
0	0.99	1.00	1.00	2348
1	1.00	0.99	0.99	2142
accuracy			0.99	4490
macro avg	0.99	0.99	0.99	4490
weighted avg	0.99	0.99	0.99	4490

Models Evaluation Results on Validation Set

Extra Trees best model evaluating...

	precision	recall	f1-score	support
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0	0.99	0.99	0.99	2348
---	------	------	------	------

1	0.99	0.99	0.99	2142
---	------	------	------	------

accuracy			0.99	4490
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macro avg	0.99	0.99	0.99	4490
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weighted avg	0.99	0.99	0.99	4490
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Models Evaluation Results on Validation Set

HistGB best model evaluating...

	precision	recall	f1-score	support
0	0.99	0.99	0.99	2348
1	0.99	0.99	0.99	2142
accuracy			0.99	4490
macro avg	0.99	0.99	0.99	4490
weighted avg	0.99	0.99	0.99	4490

Models Evaluation Results on Validation Set

MLP best model evaluating...

	precision	recall	f1-score	support
0	1.00	1.00	1.00	2348
1	1.00	1.00	1.00	2142
accuracy			1.00	4490
macro avg	1.00	1.00	1.00	4490
weighted avg	1.00	1.00	1.00	4490



OVERFITTING!



BUT...

MLP Result on Test Set

MLP model testing...

	precision	recall	f1-score	support
0	1.00	1.00	1.00	2348
1	1.00	1.00	1.00	2142
accuracy			1.00	4490
macro avg	1.00	1.00	1.00	4490
weighted avg	1.00	1.00	1.00	4490

How We Achieved This Accuracy?

All models achieved over 99% accuracy on validation set and MLP on test set, likely due to high data separability.

Strong feature engineering and powerful embeddings—especially BERT—enabled effective class distinction.

This suggests the dataset was well-structured and the models leveraged it with high representational precision.

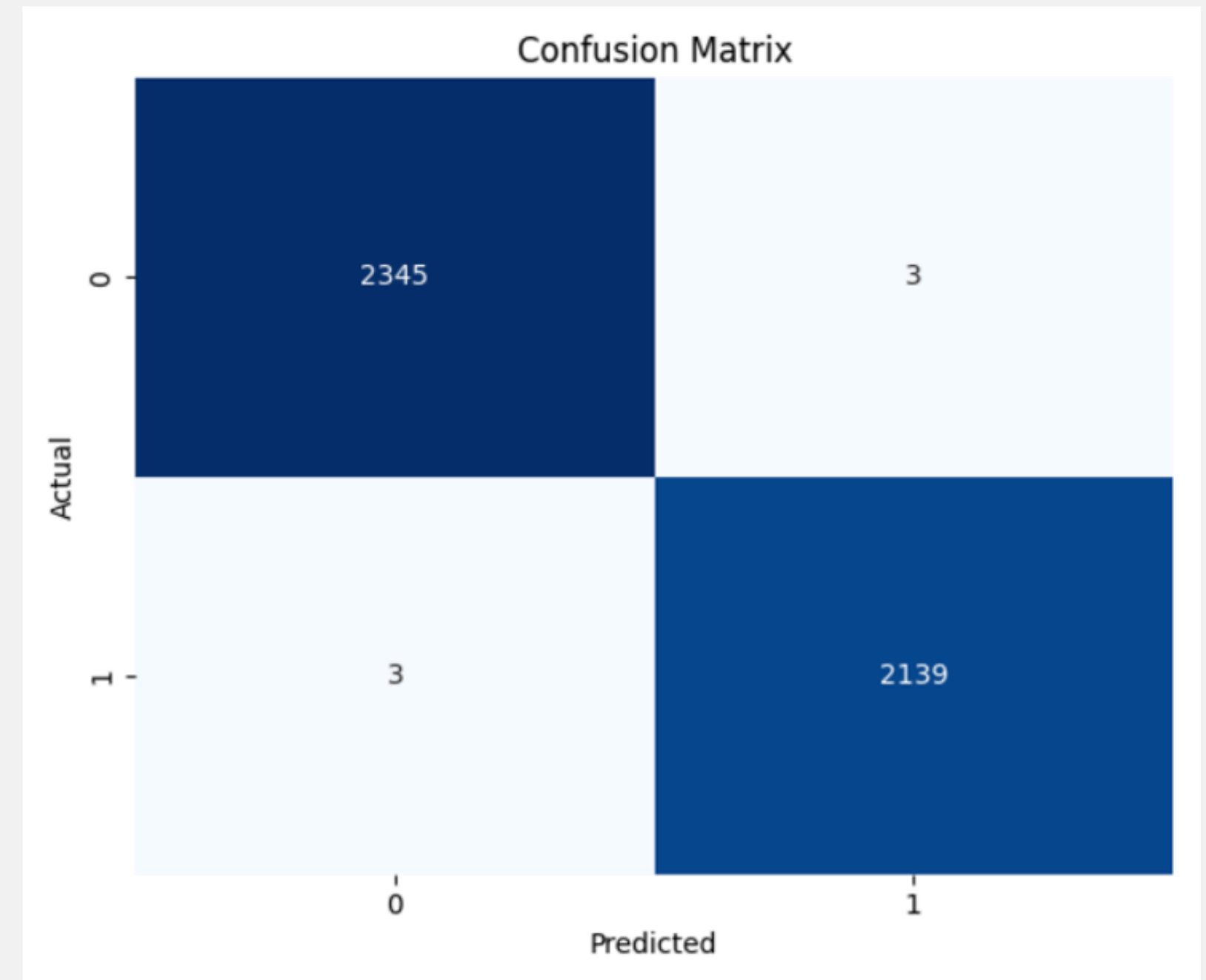
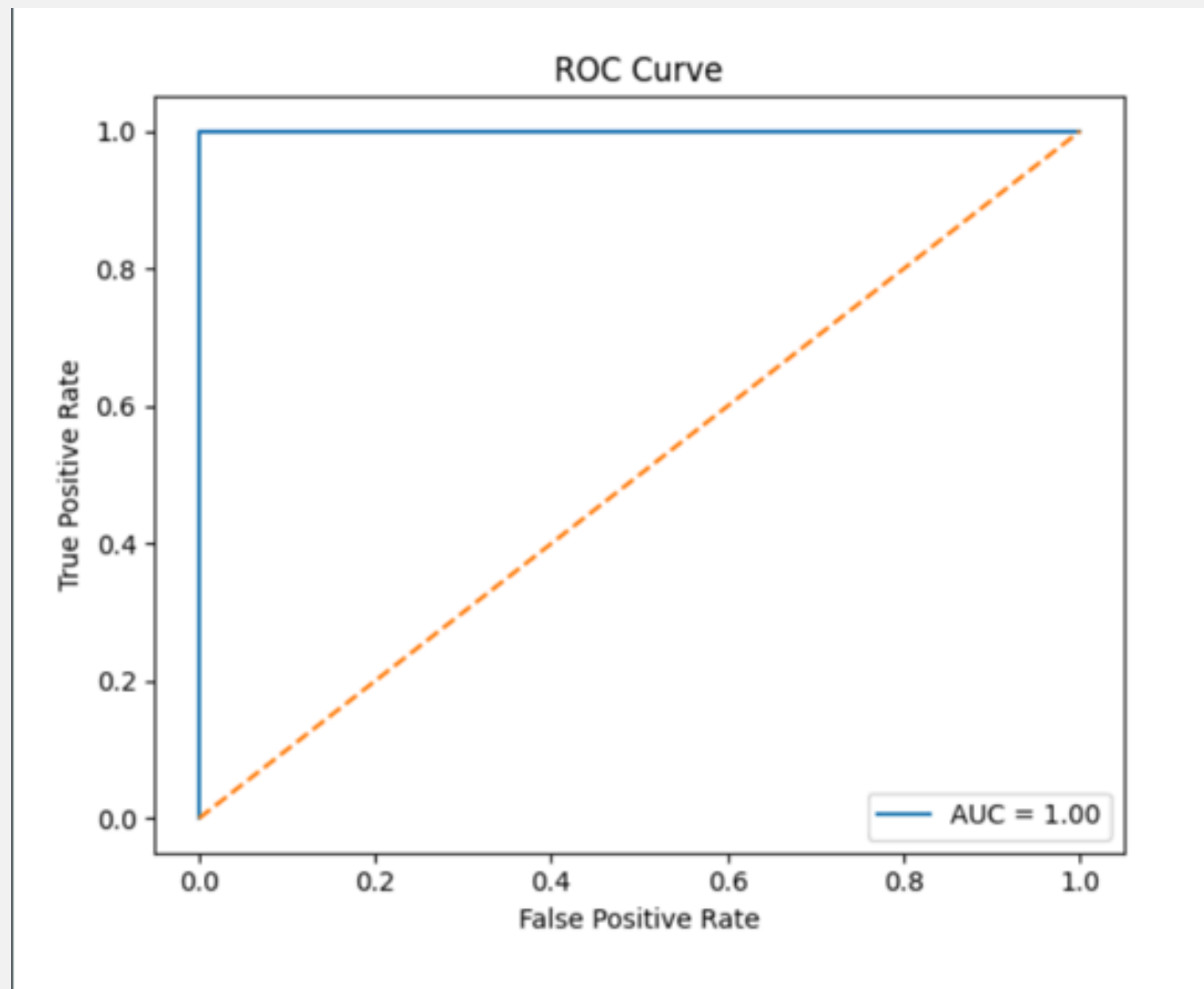


MLP Results

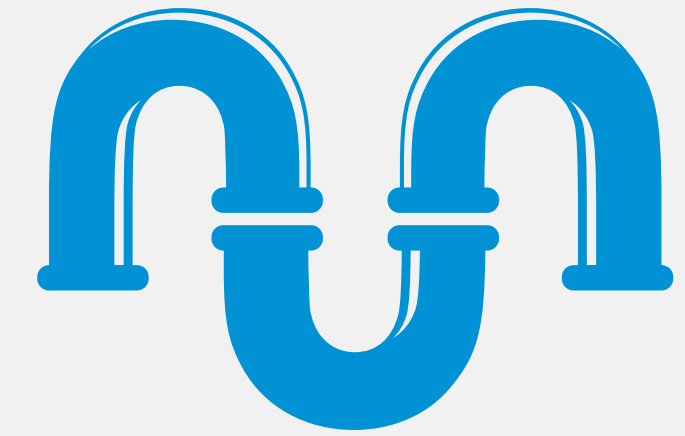
Q Search metrics	
Metric	Value
train_accuracy	0.9986636971046771
train_AUC-ROC	0.9999942339229844
train_balanced_accuracy	0.9986608783206639
train_Cohens_Kappa	0.9973217566413277
train_f1	0.9985994397759104
train_log_loss	0.003419562295541943
train_MCC	0.9973217566413277
train_precision	0.9985994397759104
train_recall	0.9985994397759104

Q Search parameters	
Parameter	Value
activation	relu
hidden_layer_sizes	(64,)
model_type	MLP

MLP Results



Pipeline



 dataset

 output

 scripts

 prediction_pipeline.py

 database

 mlruns

 notebooks

 Prediction Pipeline

 runID

 scaler

 Training Pipeline

 dataset

 queries

 scripts

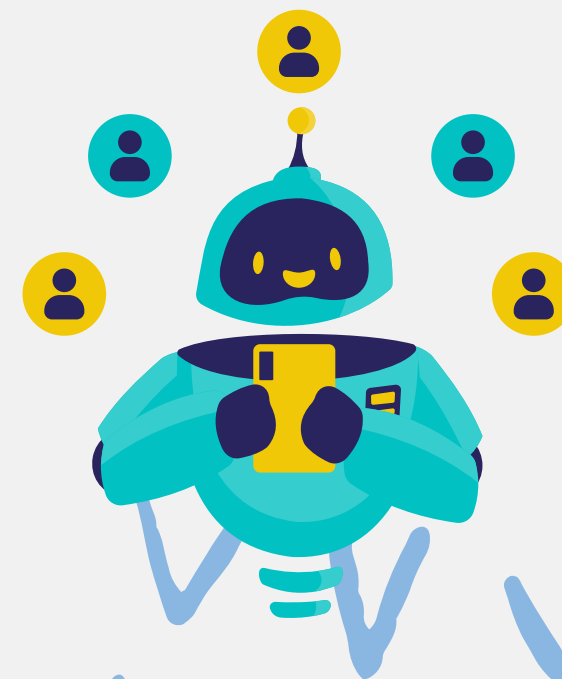
 training_pipeline.py

Fake News Detection

This project revealed that **fake news** tends to be *shorter, more emotional, and attention-seeking*, often published at the *beginning of the week* when people are more active online.

In contrast, **true news** demonstrates a *more structured style, using balanced language and richer content*, commonly *distributed during weekdays* through official channels.

These patterns show that misinformation is not just about false content, it's about how, when, and why it's delivered. Recognizing these clues is key to building awareness and resilience in the age of information warfare.





**Thank you
very much!**